

Exam July 8, 2020.

Complete solutions of selected problems.

1. There exists a linear homogeneous differential equation with constant coefficients of order 7 that has as solutions the functions: (a) $t^2 \cos 2t$, $t \sin 2t$ and e^{7t} ? ; (b) $t^2 \cos 2t$, $t \sin 7t$ and e^t ? ; (c) $t^2 \cos 2t$? . justify the answers.

Solution

we have that:

$t^2 \cos 2t$ is a sol. of a LHDE with CC iff $\pm 2i$ is a root of order (multiplicity) 3 of the charact. eq.
 $t \sin 2t$ is a sol. of a LHDE with CC iff $\pm 2i$ is a root of multiplicity 2 of the charact. eq.
 e^{7t} is a sol. of a LHDE with CC iff 7 is a root of mult. 1 (simple root) of the charact. eq.
 e^t is a sol. of a LHDE with CC iff 1 is a simple root of the charact. eq.

(a) These functions are solutions of a LHDE with CC iff $\pm 2i$ is a root of multiplicity 3 and 7 is a simple root of the charact. eq. Thus, the charact. eq. has at least degree $3 \times 2 + 1 = 7$.

Conclusion: the answer at (a) is

"YES".

(b) These functions are solutions of a LHDE with CC iff $\pm 2i$ is a root of multiplicity 3, $\pm 7i$ is a root of multiplicity 2, and 1 is a simple root of the charact. eq. Thus, the charact. eq. has at least degree $3 \times 2 + 2 \times 2 + 1 = 11$.

Conclusion: the answer at (b) is "No".

(c) Here the charact. eq. has at least degree 6.

Conclusion: the answer at (c) is "YES".

2. (a) Does the formula $x = A_0 \cos(3t - \varphi_0)$, $A_0 \geq 0$, $\varphi_0 \in [0, 2\pi]$ describe the general solution of the differential equation $x'' + 9x = 0$?

Solution First we apply the characteristic eq. method to find the general solution of $x'' + 9x = 0$.

$$r^2 + 9 = 0 \quad r_{1,2} = \pm 3i \mapsto \cos 3t, \sin 3t$$

$$x = c_1 \cos 3t + c_2 \sin 3t, \quad c_1, c_2 \in \mathbb{R}.$$

Now we work with the given formula.

$$x = A_0 \cos(3t - \varphi_0) = A_0 \cos 3t \cos \varphi_0 + A_0 \sin 3t \sin \varphi_0 =$$

$$= (A_0 \cos \varphi_0) \cos 3t + (A_0 \sin \varphi_0) \sin 3t$$

We know that $\forall (c_1, c_2) \in \mathbb{R}^2 \setminus \{0, 0\} \exists! (A_0, \varphi_0) \in (0, \infty) \times [0, 2\pi]$ such that $c_1 = A_0 \cos \varphi_0$ and $c_2 = A_0 \sin \varphi_0$.

(A_0, φ_0 are the polar coordinates of (c_1, c_2)).

Of course, for $(c_1, c_2) = (0, 0)$ we can take $A_0 = 0$.

Conclusion: The answer is "YES".

2.(b) Let $A_1 \in \mathbb{R}$. Find a particular solution of $x'' + 9x = A_1 \cos 3t$, knowing that it has the form $x_p = at \sin 3t$.

Solution. $x'_p = a \sin 3t + 3at \cos 3t$
 $x''_p = 3a \cos 3t + 3a \cos 3t - 9at \sin 3t$

we replace in the DE:

$$6a \cos 3t - 9at \sin 3t + 9at \sin 3t = A_1 \cos 3t, \quad \forall t \in \mathbb{R}.$$

$$6a \cos 3t = A_1 \cos 3t, \quad \forall t \in \mathbb{R}$$

$$\Leftrightarrow 6a = A_1 \quad \Leftrightarrow a = \frac{A_1}{6}$$

conclusion: $x_p = \frac{A_1}{6} t \sin 3t.$

2.(c) Find the solution of the IVP $\begin{cases} x'' + 9x = A_1 \cos 3t \\ x(0) = 0 \\ x'(0) = 0 \end{cases}$.

Solution.
 Let $\theta(t) = \frac{A_1}{6} t \sin 3t$. We know that θ is a sol. of the D.E. (from (b)). It is easy to check that θ satisfies also the ic's. Thus, θ is a solution of this IVP.
 We know from the lecture that this IVP has a unique sol.
 Conclusion: The unique solution of this IVP is $\theta(t) = \frac{A_1}{6} t \sin 3t$.

Describe the motion of a simple pendulum in the case that $\theta(t)$ is the measure in radians of the angle between the rod and the vertical.

Solution.
 we have that $\theta(0) = 0$ and $\theta'(0) = 0$. Thus, the critical position of the pendulum is

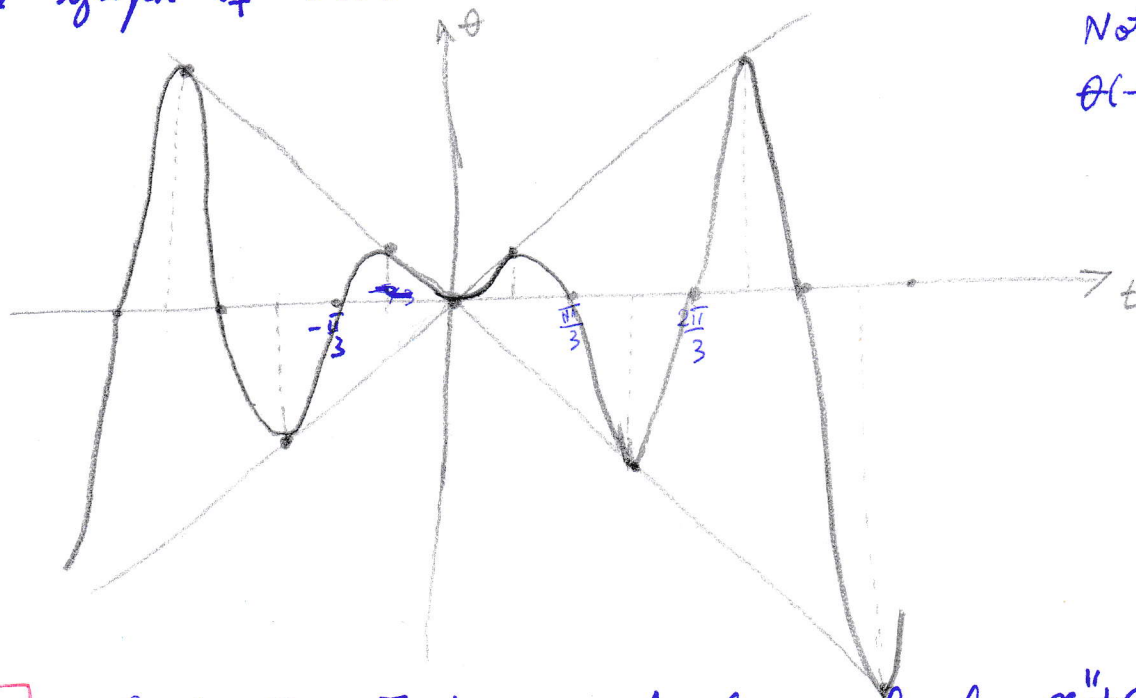


we also have that $\theta(\frac{k\pi}{3}) = 0 \quad \forall k \in \mathbb{Z}$ and θ changes sign around $\frac{k\pi}{3}$. Thus, the pendulum oscillates around the position $\theta^* = 0$.

Since $\lim_{t \rightarrow \infty} |\theta(t)| = \lim_{t \rightarrow \infty} \left| \frac{A_1}{6} t \sin 3t \right| = +\infty$

we deduce that the amplitude of the oscillations is increasing to ∞ as $t \rightarrow \infty$. Of course, practically, this is not possible. This phenomenon is called Resonance.

The graph of $\theta(t)$ looks like (fix $A_1 = 6$)



Note that
 $\theta(-t) = -\theta(t)$
 $\forall t \geq 0$

2. (d) Let $A_2 \in \mathbb{R}$. Find a particular sol. of $x'' + gx = A_2$.

Solution. $x_p = \frac{A_2}{g}$ (constant function).

2. (e) Find a particular sol. of $x'' + gx = A_1 \cos 3t + A_2$.

Solution. We know that $x_{p_1} = \frac{A_1}{6} t \sin 3t$ is a sol. of

$x'' + gx = A_1 \cos 3t$ and that $x_{p_2} = \frac{A_2}{g}$ is of $x'' + gx = A_2$.

We apply the Superposition Principle and deduce that

$$x_p = \frac{A_1}{6} t \sin 3t + \frac{A_2}{g}.$$

3.(a) Find the solution of the IVP $y' = y$, $y(0) = 1$.

Sol: $\varphi(x) = e^x$.

3.(b) Write the Euler's numerical formula with stepsize $h = 0.01$ to approximate the solution of this IVP in the interval $[0, 1]$.

Solution. $x_k = kh = k \cdot 0.01$, $k = \overline{0, 100}$

$x_0 = 0$
 $y_0 = 1$
 $y_{k+1} = y_k + h y_k \Leftrightarrow y_{k+1} = 1.01 \cdot y_k$
 $k = \overline{0, 100}$.

3.(c) Using (b) find a rational approximation of the Euler's constant e .

Solution. Since $x_{100} = 1$ and $\varphi(1) = e$ we have that y_{100} is the approximation given by (b) for e .

$y_{k+1} = 1.01 \cdot y_k$, $y_0 = 1 \Leftrightarrow y_k = (1.01)^k$, $+k \geq 0$
thus, $y_{100} = \left(\frac{101}{100}\right)^{100}$ which is a rational number.

4. (a) We consider the planar system

$$\begin{cases} \dot{x} = -y\sqrt{3} + x(9-x^2-3y^2) \\ \dot{y} = \frac{x}{\sqrt{3}} + y(9-x^2-3y^2) \end{cases}$$

(a) Study the type and stability of the equilibrium point $(0,0)$ using the linearization method.

Solution. First note that $(0,0)$ is, indeed, an equil. point.

Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}^2$, $f(x,y) = \begin{pmatrix} -y\sqrt{3} + 9x - x^3 - 3xy^2 \\ \frac{x}{\sqrt{3}} + 9y - x^2y - 3y^3 \end{pmatrix}$.

f is a polynomial function, thus $f \in C^1(\mathbb{R}^2)$.

$$Jf(x,y) = \begin{pmatrix} 9-3x^2-3y^2 & -\sqrt{3}-6xy \\ \frac{1}{\sqrt{3}}-2xy & 9-x^2-9y^2 \end{pmatrix}$$

$$Jf(0,0) = \begin{pmatrix} 9 & -\sqrt{3} \\ \frac{1}{\sqrt{3}} & 9 \end{pmatrix} \quad \begin{vmatrix} 9-\lambda & -\sqrt{3} \\ \frac{1}{\sqrt{3}} & 9-\lambda \end{vmatrix} = 0 \Leftrightarrow$$

$$\Leftrightarrow (9-\lambda)^2 + 1 = 0 \Leftrightarrow (9-\lambda)^2 = -1 \Leftrightarrow 9-\lambda = \pm i$$

$$\Leftrightarrow \lambda_{1,2} = 9 \pm i$$

$\text{Re}(\lambda_1) = \text{Re}(\lambda_2) = 9 \neq 0 \Rightarrow$ the eq. $(0,0)$ is hyperbolic $\Rightarrow$

$\Rightarrow$ the linearization method works

The linearized system around $(0,0)$ is $\begin{cases} \dot{x} = 9x - y\sqrt{3} \\ \dot{y} = \frac{x}{\sqrt{3}} + 9y \end{cases}$

which is of focus type since the eigenvalues are complex with non-zero real part. It is also a global repeller since $\text{Re}(\lambda_1) = \text{Re}(\lambda_2) > 0$.

Applying the linearization method, we deduce that the eq. $(0,0)$ of the given nonlinear system is a repeller.

4.(b) Check that $\varphi(t, 3, 0) = (3 \cos t, \sqrt{3} \sin t)$, $\forall t \in \mathbb{R}$.

Represent the corresponding orbit. What shape is it?

Solution By definition, $\varphi(t, 3, 0)$ denotes the solution of the system that also satisfies the initial conditions $x(0) = 3$, $y(0) = 0$.

Let $x(t) = 3 \cos t$ and $y(t) = \sqrt{3} \sin t$. It is easy to check that, indeed, they satisfy the ~~given~~ initial cond. above. Now we check that ~~they~~ ~~(x(t), y(t))~~ they satisfy the system.

$$\begin{aligned} -3 \sin t &= -\sqrt{3} \cdot \sqrt{3} \sin t + 3 \cos t (9 - 9 \cos^2 t - 3 \cdot 3 \sin^2 t) \\ \sqrt{3} \cos t &= \frac{3}{\sqrt{3}} \cos t + \sqrt{3} \sin t (9 - 9 \cos^2 t - 3 \cdot 3 \sin^2 t) \end{aligned} \quad \forall t \in \mathbb{R}.$$

It is easy to see that this is valid.

Conclusion: $\varphi(t, 3, 0) = (3 \cos t, \sqrt{3} \sin t)$, $\forall t \in \mathbb{R}$.

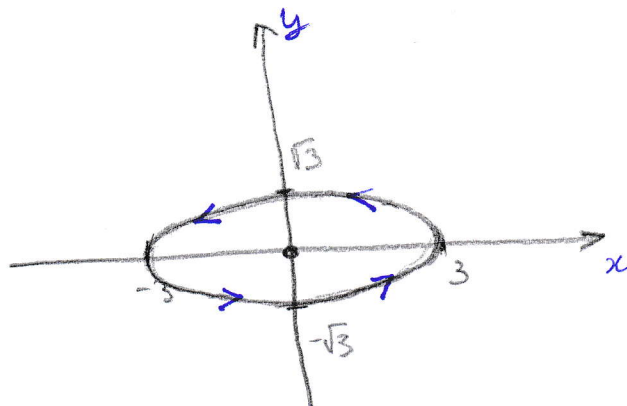
By definition, the corresponding orbit is

$$\Gamma = \left\{ \underbrace{(3 \cos t)}_x, \underbrace{(\sqrt{3} \sin t)}_y : t \in \mathbb{R} \right\}.$$

we have :

$$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{\sqrt{3}}\right)^2 = 1$$

we know that this is the implicit equation of an ellipse. Thus, Γ is an ellipse. Its representation is :



$$t=0 \quad x(0)=3, y(0)=0$$

$$t=\frac{\pi}{2} \quad x\left(\frac{\pi}{2}\right)=0, y\left(\frac{\pi}{2}\right)=\sqrt{3}$$

$$(3,0) \rightarrow (0,\sqrt{3})$$

$$t=0 \rightarrow t=\frac{\pi}{2}$$

4. c) Transform the given system to the coordinates $(r, \varphi) \in [0, \infty) \times [0, 2\pi)$ related to the Cartesian coordinates (x, y) by $\frac{x}{\sqrt{3}} = r \cos \varphi, y = r \sin \varphi$.

Solution. we have $\left(\frac{x}{\sqrt{3}}\right)^2 + y^2 = r^2, \quad \operatorname{tg} \varphi = y \cdot \frac{\sqrt{3}}{x}$

$$\Leftrightarrow \begin{cases} x^2 + 3y^2 = 3r^2 \\ \operatorname{tg} \varphi = \sqrt{3} \cdot \frac{y}{x} \end{cases} \quad \text{Derivating w.r.t. } t \text{ we obtain:}$$

$$3r\dot{r} = x\dot{x} + 3y\dot{y}$$

$$\frac{\dot{\varphi}}{\cos^2 \varphi} = \sqrt{3} \cdot \frac{\dot{y}x - \dot{x}y}{x^2}$$

Then:

$$\begin{cases} 3r\dot{r} = x[-y\sqrt{3} + x(9-x^2-3y^2)] + 3y\left[\frac{x}{\sqrt{3}} + y(9-x^2-3y^2)\right] \\ \frac{\dot{\varphi}}{\cos^2 \varphi} = \sqrt{3} \cdot \frac{x\left[\frac{x}{\sqrt{3}} + y(9-x^2-3y^2)\right] - y[-y\sqrt{3} + x(9-x^2-3y^2)]}{x^2} \end{cases}$$

$$\Leftrightarrow \begin{cases} 3r\dot{r} = (x^2 + 3y^2)(9 - x^2 - 3y^2) \\ \frac{\dot{\varphi}}{\cos^2 \varphi} = \sqrt{3} \cdot \frac{\frac{x^2}{\sqrt{3}} + y^2\sqrt{3}}{x^2} = \frac{x^2 + 3y^2}{x^2} \end{cases} \quad (\Leftrightarrow)$$

$$\Leftrightarrow \begin{cases} 3r\dot{r} = 3r^2(9 - 3r^2) \\ \frac{\dot{\varphi}}{\cos^2 \varphi} = \frac{3r^2}{3r^2 \cos^2 \varphi} \end{cases} \quad (\Leftrightarrow) \quad \begin{cases} \dot{r} = r(9 - 3r^2) \\ \dot{\varphi} = 1 \end{cases}$$

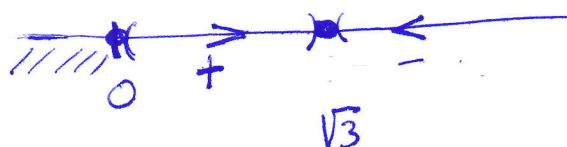
4. (d) Sketch the phase portrait of this planar system.

In the coordinates (r, φ) we obtained ~~an~~ ^{the} uncoupled system

$$\begin{cases} \dot{r} = r(9 - 3r^2) \\ \dot{\varphi} = 1 \end{cases}$$

We represent first the phase portrait of each equation.

$$\dot{r} = r(3 + r\sqrt{3})(3 - r\sqrt{3})$$



$$\dot{\varphi} = 1$$



Note that the new coordinates (r, φ) are very much like the polar coordinates, just a little bit deformed. r is like a distance to the origin, and φ is like an angle of the vector (x, y) with the Ox semi-axis. Using the geometric interpretation of polar coordinates and taking into account the particularities of the new coordinates we deduce:

- since the angle φ is strictly increasing, any orbit will rotate around the origin in the trigonometric sense.
- the equilibrium point $r^* = \sqrt{3}$ of $\dot{r} = r(9 - 3r^2)$ corresponds to the orbit Γ found at (b).
 a point with $r \in (0, \sqrt{3})$ lies inside Γ , while
 a point with $r > \sqrt{3}$ lies outside Γ .

- reading the phase portrait of $\dot{r} = r(9-3r^2)$ and taking into account the above remarks, we deduce that an orbit that starts inside Γ , ~~also~~ stays inside Γ for all ~~to~~ any time and is departing from the origin. ~~Also, an orbit that starts~~ and approaching Γ as $t \rightarrow \infty$. Also, an orbit that starts outside Γ , stays outside Γ for all time and is approaching Γ as $t \rightarrow \infty$. Thus, the phase portrait looks like

